

Dengue, Fais Island, Micronesia, 2011

Overview

This dataset describes a dengue fever outbreak on Fais Island, a small island in Yap State, Federated States of Micronesia, in 2011. The outbreak affected 157 of the island's 294 residents (53% attack rate), illustrating the rapid spread of dengue in a small, immunologically naïve island population.

Source: Funk S, Kucharski AJ, Camacho A, Eggo RM, Yakob L, Murray LM, Edmunds WJ (2016). Comparative analysis of dengue and Zika outbreaks reveals differences by setting and virus. *PLOS Neglected Tropical Diseases* 10(12):e0005173.

Loading the data

```
using DataFrames
using Outbreaks
df = Outbreaks.dengue_fais_2011()
size(df)
```

(57, 3)

First few rows

```
first(df, 6)
```

	onset_date	nr	value
	Date	Int64	Int64
1	2011-09-15	7	0
2	2011-09-22	14	0
3	2011-09-29	21	0
4	2011-10-06	28	0
5	2011-10-13	35	0
6	2011-10-20	42	0

Summary statistics

```
describe(df)
```

	variable	mean	min	median	max	nmissing	eltype
	Symbol	Union...	Any	Any	Any	Int64	DataType
1	onset_date		2011-09-15	2011-11-30	2012-02-14	0	Date
2	nr	83.0	7	83.0	159	0	Int64
3	value	2.7193	0	0.0	19	0	Int64

Epidemic curve

```
sort!(df, :onset_date)
println("Onset range: $(minimum(df.onset_date)) to $(maximum(df.onset_date))")
println("Total value: $(sum(df.value))")
for row in eachrow(df)
```

```
if row.value > 0
  bar = repeat("█", min(row.value, 80))
  println("${row.onset_date} (wk ${row.nr}) | $bar ${row.value}")
end
end
```

Onset range: 2011-09-15 to 2012-02-14

Total value: 155

